

Global Maritime Route Optimization System

VRP-Based Decision Support & Simulation

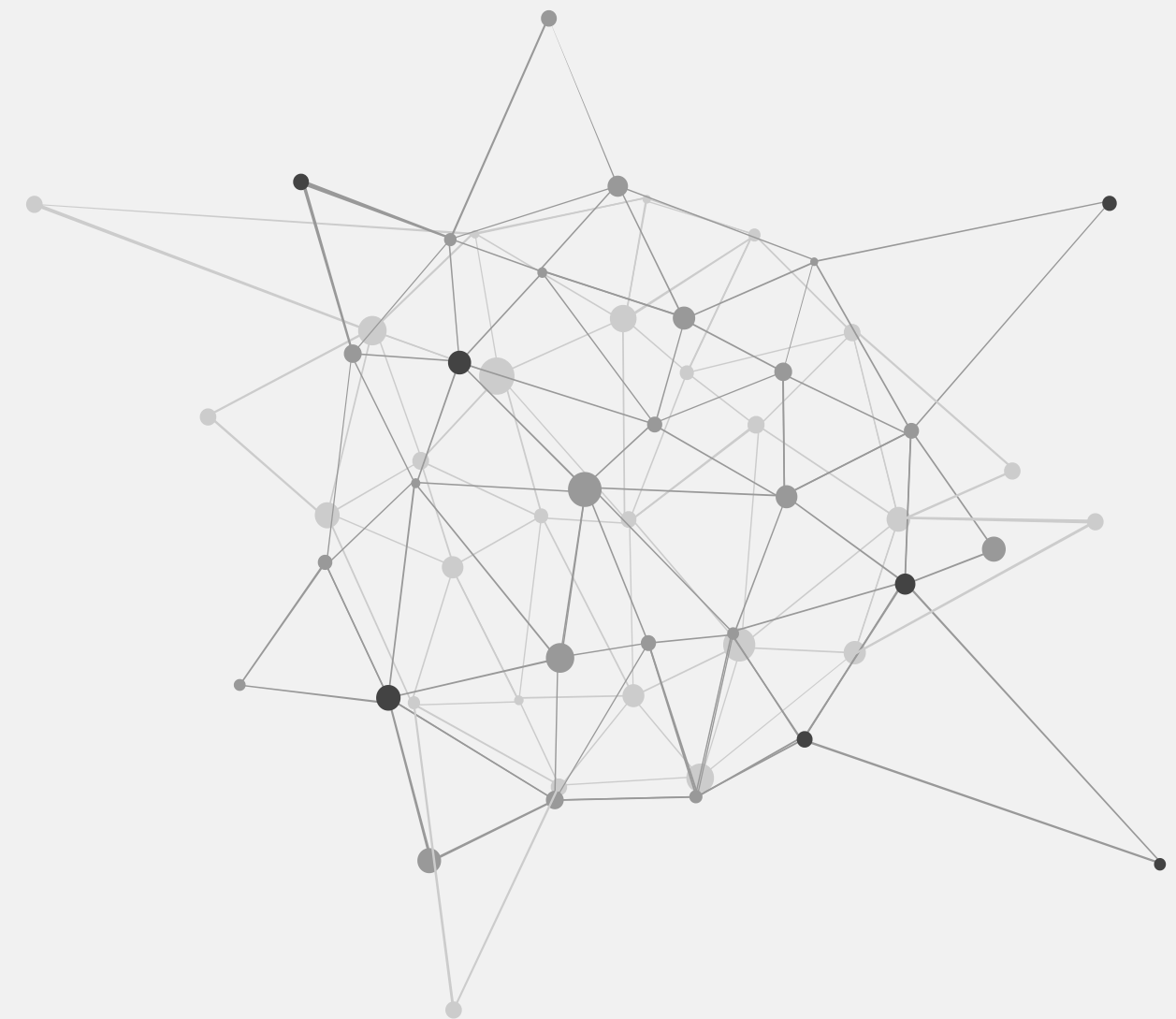


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The NP-Hard Challenge of Global Fleet Assignment

Traditional heuristic algorithms collapse under the massive search spaces of modern shipping networks.

Our Objective: Designing a custom "Hybrid Genetic Algorithm and VNS" from scratch to minimize operational costs, rather than relying on the traditional heuristic algorithms presented in the literature.



Mathematical Model and Operational Constraints

Objective Function: Minimize (Z) = Route Cost + Operating Expenses + Inefficiency Penalties – Charter Revenues

01.



35-Day Cycle

The total voyage duration of a vessel must be restricted to a window of 33–37 days.

02.



7-Day Leg Limit

The maximum travel time between two consecutive stops must not exceed 7 days.

03.



Capacity (TEU)

The cargo volume transported cannot exceed the designated vessel capacity.

04.



Regional Segmentation

Routes must be segregated into Northern and Southern sectors.

05.

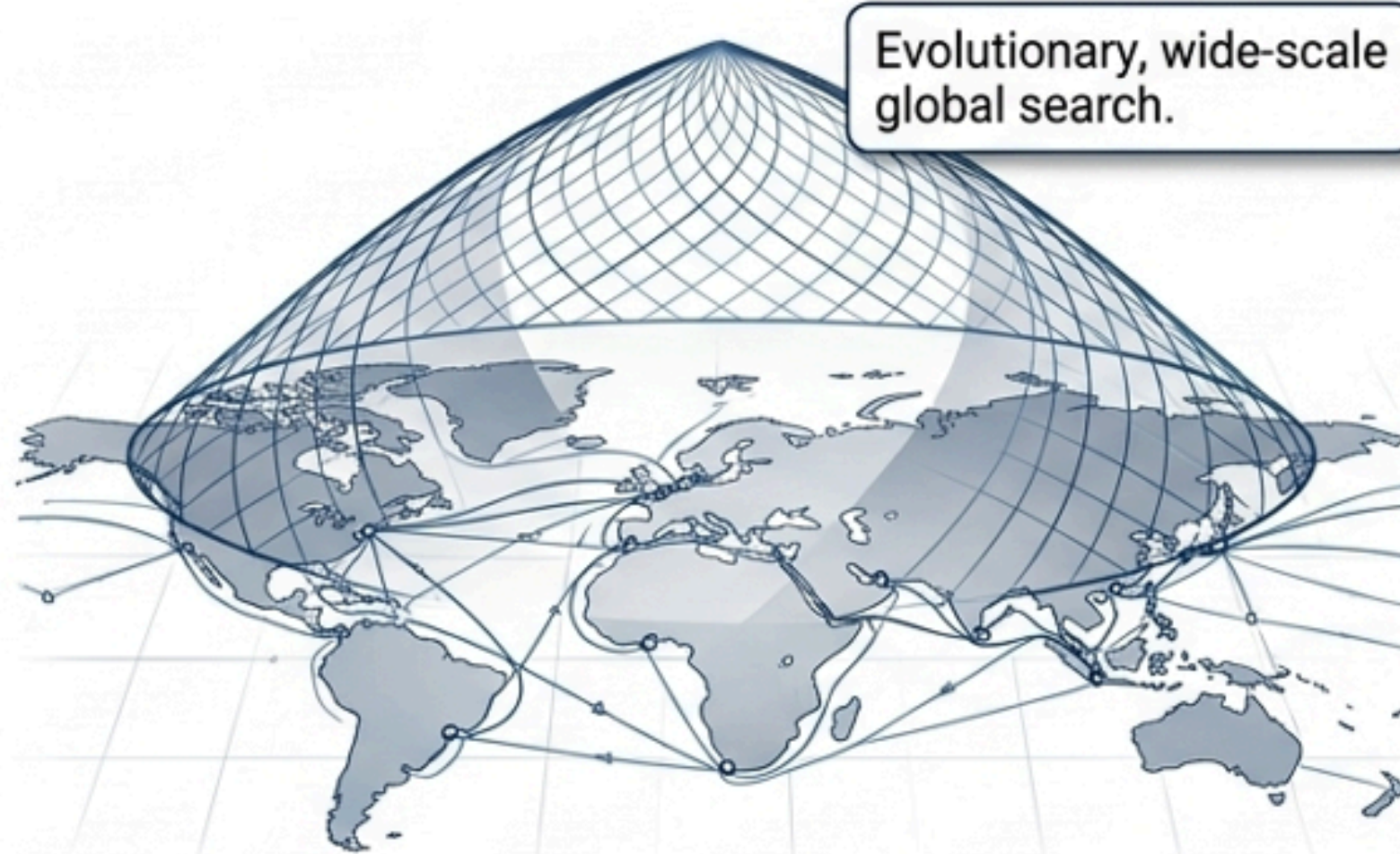


Maximum 2 Stops

A vessel is permitted to call at a maximum of 2 delivery ports.

Why Hybridize? (GA + VNS) Algorithm

Genetic Algorithm (The Macro Explorer)



Crosses different routes to rapidly navigate the massive search space and establish the core routing framework.

VNS (The Micro Refiner)



Takes the strong baseline from the GA and makes microscopic, localized improvements to squeeze out final inefficiencies.

GA builds the foundation; VNS perfects the margin.

Financial Proof: Benchmark Algorithm Performance

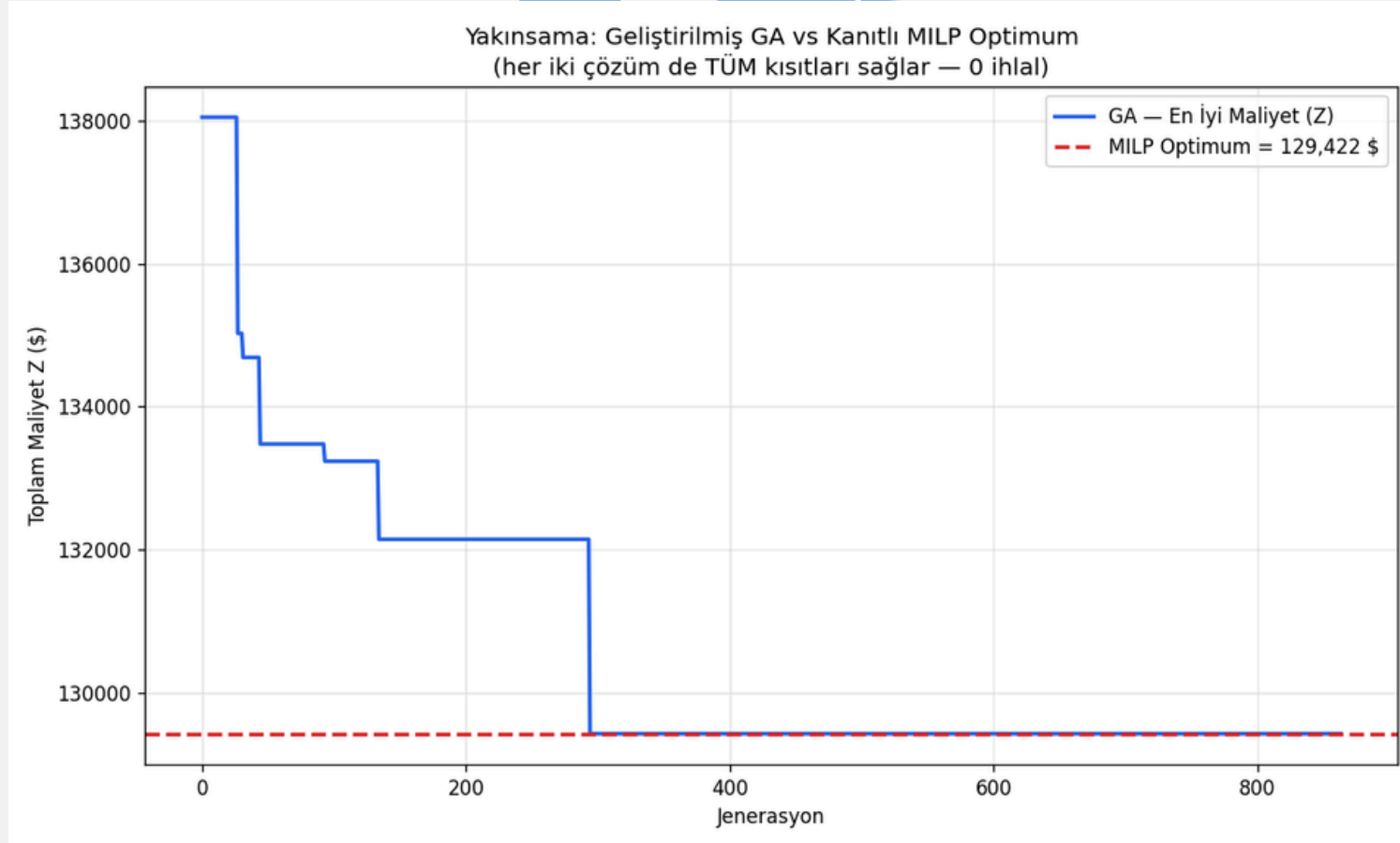
Pre-Optimization Cost: ~\$138,051 (Average randomized feasible solutions strictly satisfying all constraints).

Post-Optimization Cost: \$129,421.62 (100% Feasible, 0 constraint violations).

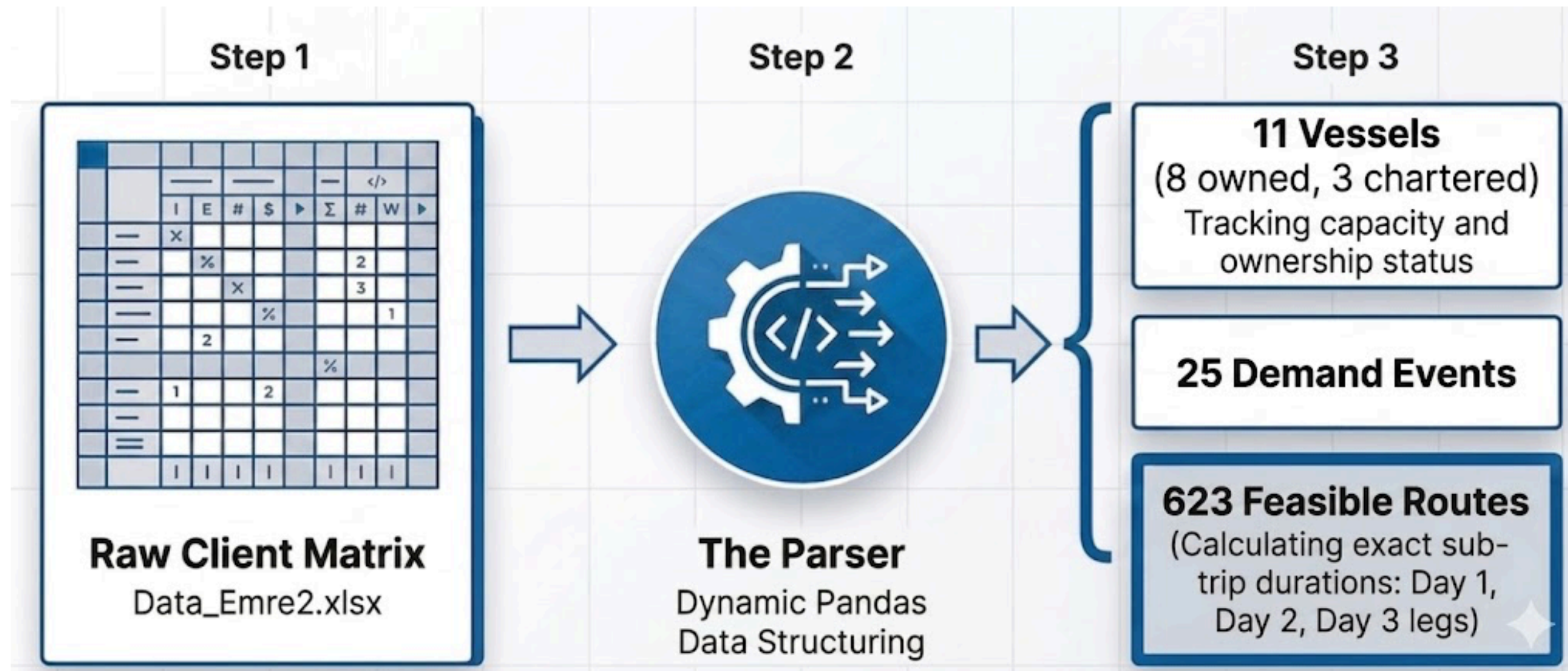
Cost Savings: ~\$8,630 (Real operational savings achieved purely from optimizing feasible voyage assignments, without artificial penalty inflations).

Optimality Proof: Results perfectly matched the exact Mixed Integer Linear Programming (MILP) global optimum with a 0.00% gap.

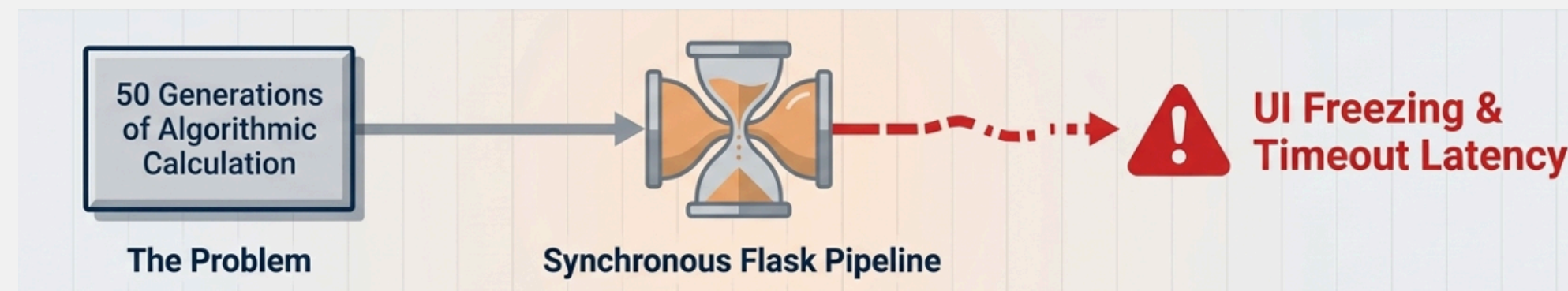
Dataset: Data_Emre2.xlsx (623 feasible candidate routes, 11 vessels, 25 demand events, Northern/Southern ports).



Establishing Data Architecture (Data Parsing)



FastAPI and Asynchronous Server



01

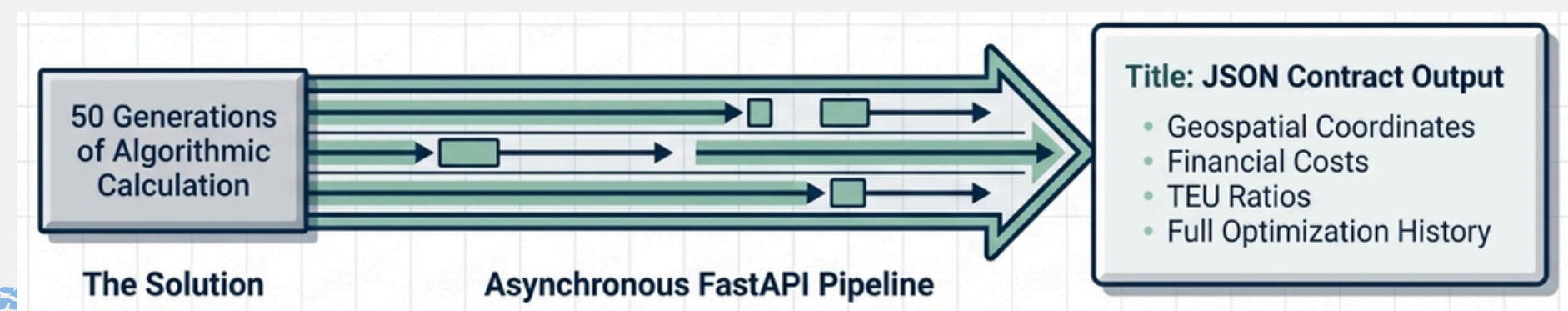
Backend Technology
FastAPI (instead of traditional frameworks like Flask).

02

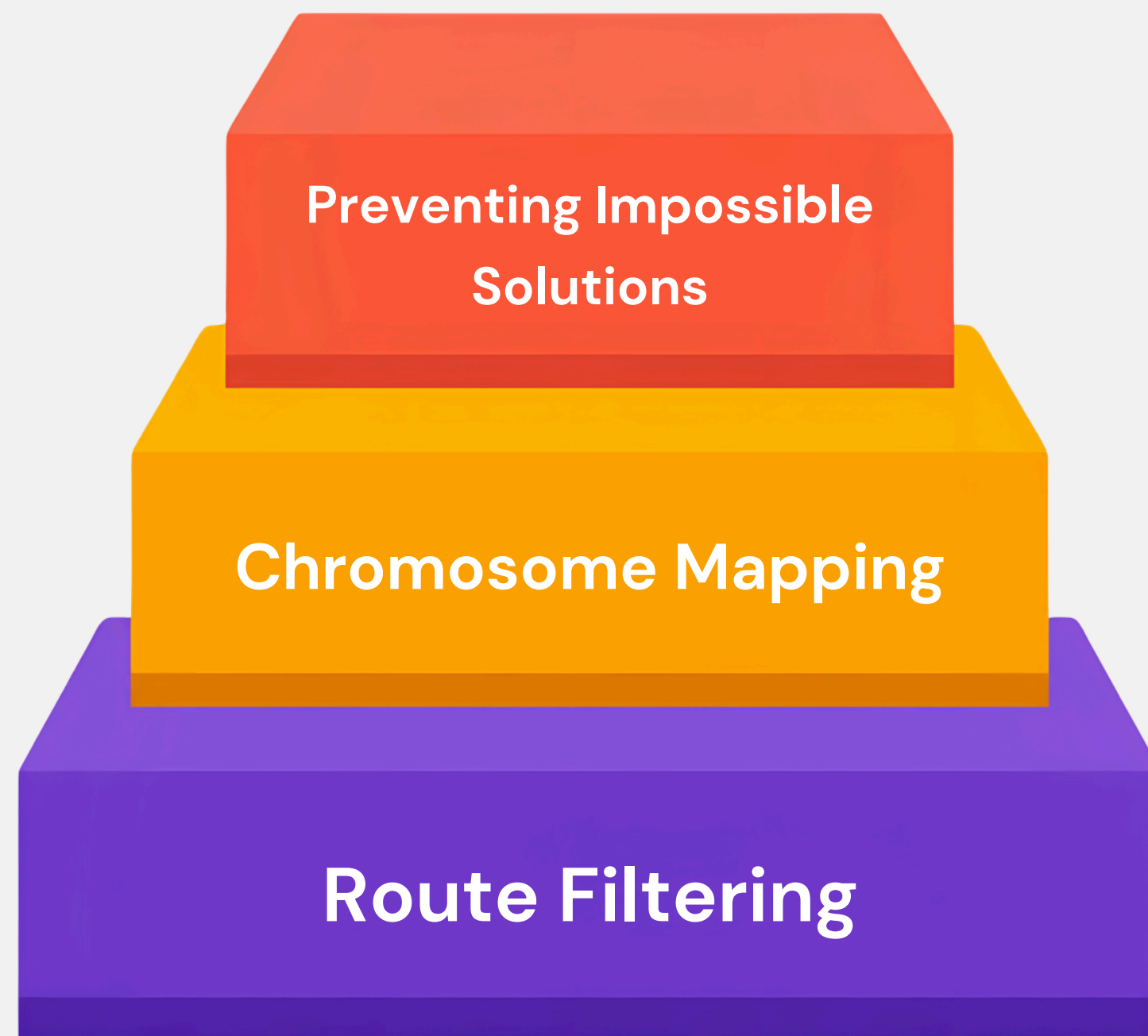
The Need for High Throughput
Calculating CPU-heavy evolutionary generations (50 iterations) in real-time.

03

API Output
A structured JSON contract delivered to the frontend in seconds, containing coordinates, calculated costs, TEU load factors, and optimization logs.



Feasibility by Construction



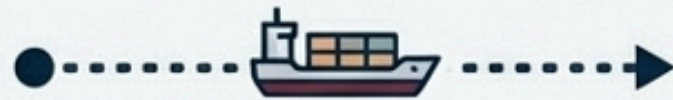
RESULT: IMPOSSIBLE SOLUTIONS ARE STRUCTURALLY PREVENTED FROM BEING GENERATED.

CHROMOSOME REDESIGNED TO MAP DEMAND EVENTS DIRECTLY TO THE OPTIMAL VOYAGES.

UNFEASIBLE ROUTES ARE PRE-FILTERED BEFORE THE SIMULATION BEGINS.

Escaping Local Optima via Variable Neighborhood Search

the ga baseline



RETRIEVE THE BEST
SOLUTION GENERATED BY
THE GA.

the relocate matrix



IMPLEMENT INCREMENTAL
RELOCATE OPERATIONS AND
ITERATED LOCAL SEARCH
(SHAKING) TO ESCAPE LOCAL
OPTIMA.

the fitness check

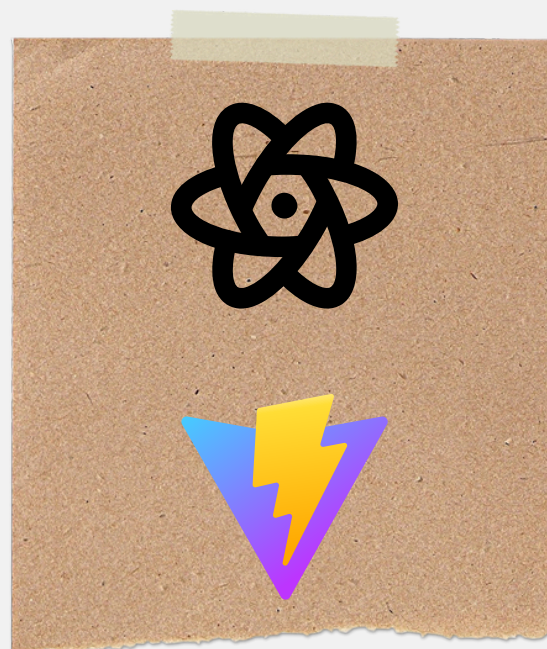


COMMIT THE RELOCATION
IF THE FITNESS VALUE
(TOTAL COST) IMPROVES.



Full Stack Visualisation

frontend



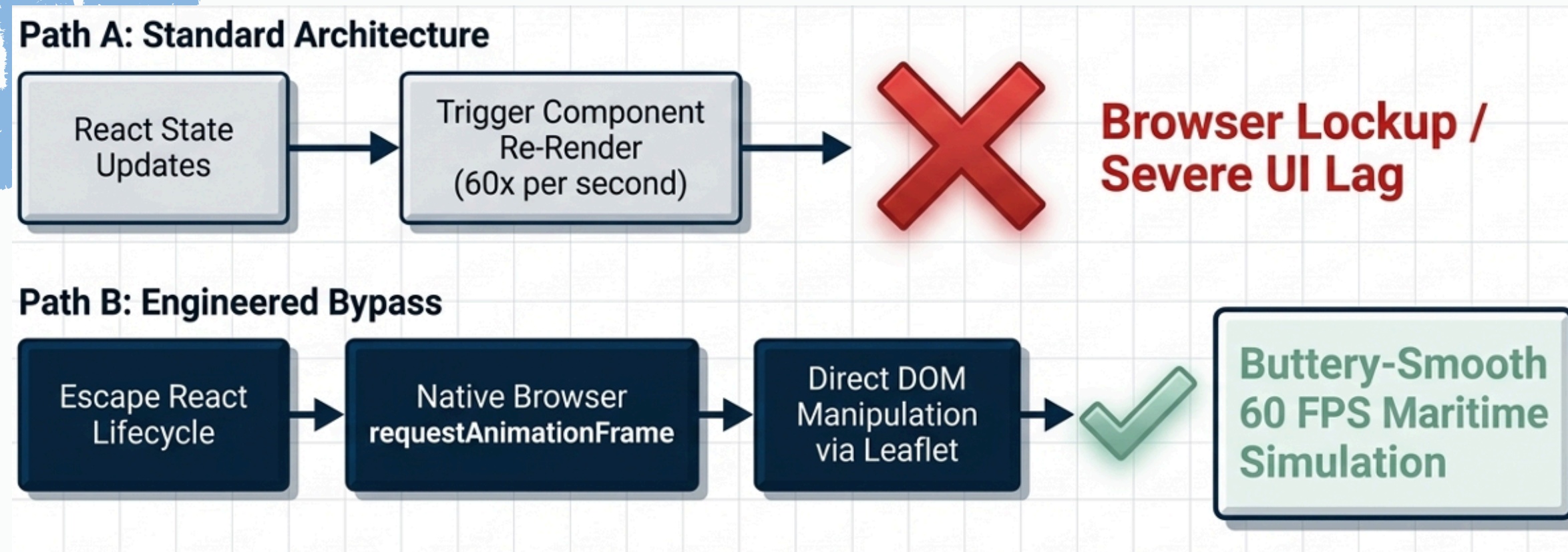
Realistic Maritime Routing
(searoute-js):

Calculating realistic oceanic channel routes around continents rather than using straight lines (as the crow flies).

Glassmorphism UI:

A premium dashboard supporting Bilingual options (TR/EN) and Dark/Light modes, designed for logistics managers.

UI Engineering: Bypassing the React Rendering Bottleneck



01

Challenge

Rendering real-time vessel movements via React state updates caused browser lag and frame drops (frequent re-rendering overhead).

02

Solution

Bypassing React's render cycle! Direct Leaflet (DOM) manipulation executed within the browser's native requestAnimationFrame loop.

03

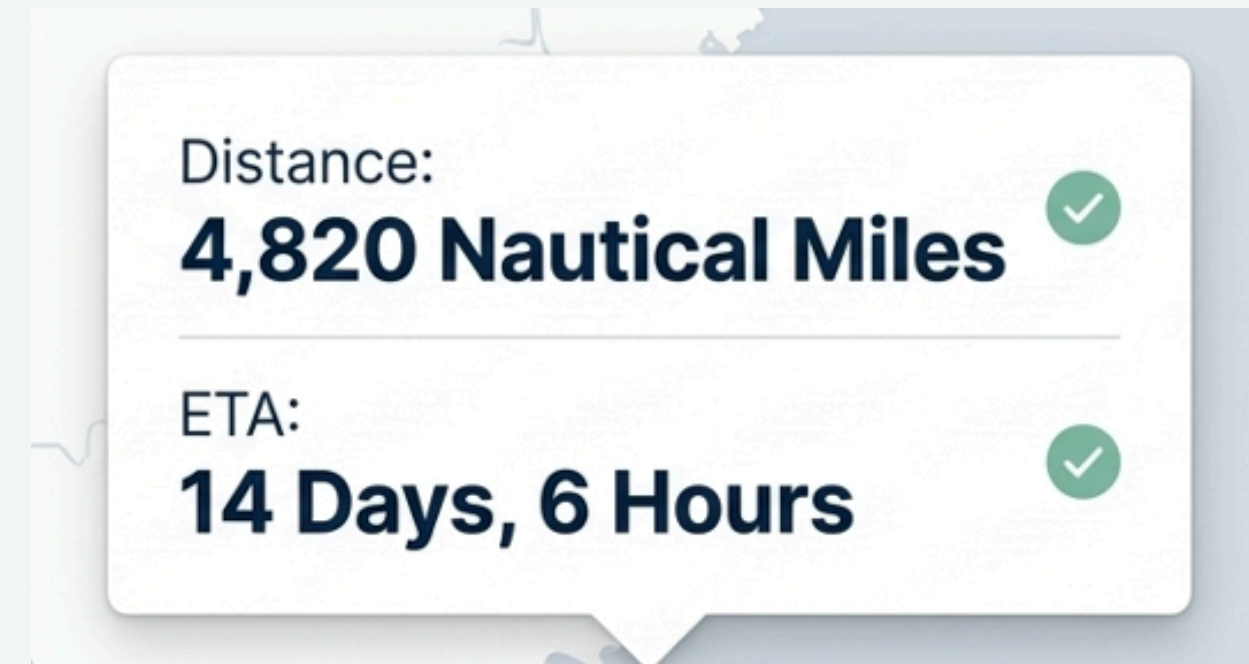
Result

A fluid, hardware-accelerated 60 Frames Per Second (FPS) simulation.

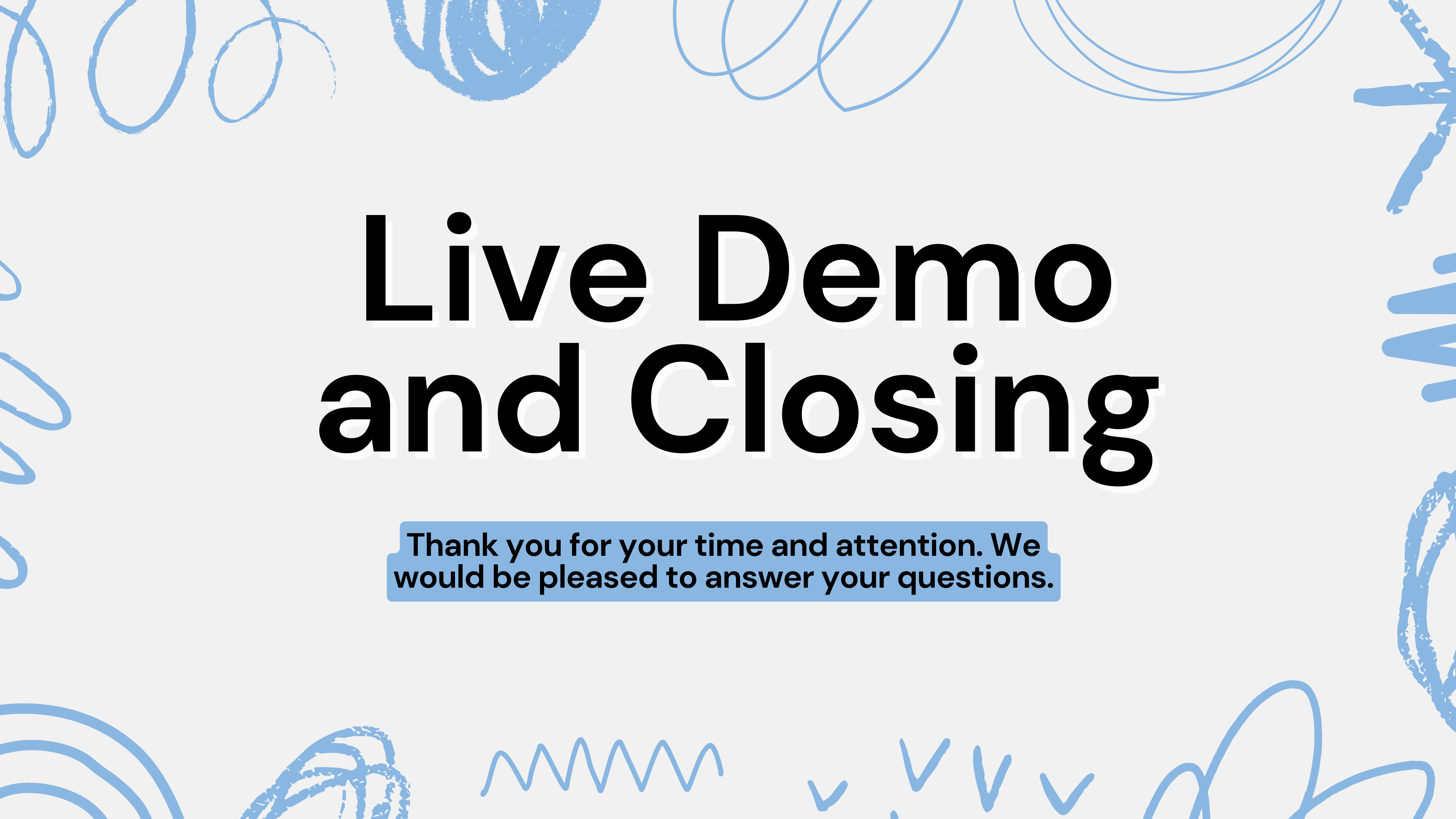
Visual Validation and Analytics



**PROVING THE EXACT MILP
VALIDATION AND 100%
FEASIBILITY WORKS
FLAWLESSLY IN PRACTICE.**



**ETA (ESTIMATED TIME OF
ARRIVAL): DYNAMICALLY
CALCULATED ARRIVAL TIMES
BASED ON SPEED AND PHYSICAL
NAUTICAL DISTANCE.**



Live Demo and Closing

Thank you for your time and attention. We
would be pleased to answer your questions.